

MFT Unit 05

1. A multi track Turing machine can described as a 6-tuple $(Q, \Sigma, \Gamma, \delta, q_0, F)$ where Σ represents:
 - a) input alphabet
 - b) tape alphabet
 - c) shift symbols
 - d) none of the mentioned
2. Which of the following is/are not true for recursively enumerable languages?
 - a) Partially decidable
 - b) Turing acceptable
 - c) Turing Recognizable
 - d) None of the mentioned
3. A Turing machine with several tapes is known as:
 - a) Multi-tape Turing machine
 - b) Poly-tape Turing Machine
 - c) Universal Turing machine
 - d) All of the mentioned
4. A multiple Turing machine is _____ powerful than a single tape Turing machine:
 - a) More
 - b) Less
 - c) None of the mentioned
5. Universal TM influenced the concept of:
 - a) Stored program computers
 - b) Interactive implementation of programming language

- c) Computability
- d) All of these

6. Which of the following are undecidable problems?

- a) Determining whether two grammars generate the same language
- b) Determining whether a grammar is ambiguous
- c) Both (a) and (b)
- d) None of the mentioned

7. Which one of the following is true for the given?

$A = \{(M, w) \mid M \text{ is a Turing machine that accepts string } w\}$

- a) A concrete undecidable problem

8. Which problem is undecidable:

- a) Finiteness problem for FSA's
- b) Membership problem for CFG's
- c) Equivalence problem for FSA's
- d) Ambiguity problem for CFG's

9. Which of the following statements are undecidable?[BL-2]

For a given Turing Machine M ,

- a) does M halt on an empty input tape
- b) does M halt for only inputs at all?
- c) is $L(M)$ regular? Context free? Turing decidable?
- d) all of the mentioned

10. Out of the three decision problems P_1 , P_2 , and P_3 , P_1 is decidable and P_2 is undecidable. The statement that holds true is:

- a) P_3 is decidable if P_1 is reducible to complement of P_2
- b) P_3 is decidable if P_1 is reducible to P_2
- c) P_3 is undecidable if P_1 is reducible to P_2
- d) P_3 is undecidable if P_2 is reducible to P_3

-
11. Define a recursively enumerable language. Under what conditions can a recursively enumerable language be classified as recursive?
 12. Define a recursive language. How does it differ from a recursively enumerable language?
 13. Write short notes on the following topics:
 - i) Turing Reducibility
 - ii) Multi-Tape Turing Machine
 14. Write Short Note on NP-completeness. Explain the concept of NP-completeness with examples of well-known NP-complete problems.
 15. Explain the Halting Problem of a Turing Machine. Why is it undecidable? Provide a basic Explanation.
 16. Describe the model of a Non-Deterministic Turing Machine (NDTM). Explain how it differs from a deterministic Turing machine. Is an NDTM more powerful than a basic Turing Machine? Justify your answer.
 17. Explain the concept of a Multi-Tape Turing Machine. How does it function, and in what way does it differ from a single tape Turing Machine? Is it more powerful than a basic Turing Machine? Justify your answer.
 18. Explain the concept of a decidable problem. When do we say that a problem is decidable? Provide an example of an undecidable problem.
 19. Give an example of a language that is not recursively enumerable. Explain why this language does not fall under the category of recursively enumerable languages.
 20. Give an example of an undecidable problem. Explain why this problem is undecidable with appropriate reasoning.

21. Explain the concepts of recursive and recursively enumerable languages.
Provide a detailed comparison between these two types of languages and illustrate with suitable examples.
22. Prove that the union of two recursively enumerable languages is also recursively enumerable. Provide a formal proof.
23. Prove that the union of two recursive languages is also recursive. Provide a logical explanation with proof.
24. Prove that the following problem is undecidable: “Given two context-free grammars (CFGs) G_1 and G_2 , determine whether the intersection of their languages, $L(G_1) \cap L(G_2)$, is empty”.
25. Prove that the Post Correspondence Problem (PCP) is undecidable.
Formally define the PCP problem and provide a proof demonstrating its undecidability.